

DRIVER SAFETY AND EMERGENCY ASSISTANCE SYSTEM

Problem Statement:

Road accidents caused by driver fatigue, distraction, and delayed emergency response continue to be a major global safety concern. Existing driver monitoring systems primarily rely on single-source data and often fail to accurately detect unsafe driving conditions in real-world environments.

To address this issue, we propose an AI-powered Intelligent Driver Safety and Emergency Assistance System that utilizes multimodal data, including video, audio, text, and sensor inputs, to continuously monitor driver behavior. The system detects signs of drowsiness, distraction, and stress in real time, provides immediate alerts to prevent accidents, and performs predictive analysis to identify potential risks. In emergency situations, it automatically shares the driver's GPS location and sends alerts to predefined contacts, enabling faster rescue response and improving overall road safety.

Dataset Annotation

The dataset used for training the driver drowsiness detection model was manually annotated using Labelme, an open-source image annotation tool. Frames were extracted from driver monitoring videos and annotated with bounding boxes around the driver's face. Each annotation was assigned one of the predefined class labels: Normal or Drowsy.

The annotated images and corresponding JSON annotation files were subsequently converted into YOLO format for model training. This annotation process ensured accurate localization and classification of driver states, enabling the development of a reliable real-time drowsiness detection system.

Annotated Dataset Link: [Annotated Data Link](#)

Annotation Tool Used: Labelme

Classes Annotated:

- Normal
- Drowsy

Annotation Format: Bounding Box Annotations